

(54) PYRAZOLOPYRIMIDINONE COMPOUNDS FOR THE TREATMENT OF X

(57) Abstract

Compounds of formula (I) and pharmaceutically acceptable salts thereof, useful as inhibitors of phosphodiesterase 5.

Examples

Representative compounds and their IC₅₀ values (nM) against PDE5 are listed in Table 1. Substituent groups R₁–R₃ follow the numbering of formula (I).

Cpd	R ₁	R ₂	R ₃	IC ₅₀ (nM)	logP
1	Me	n-Pr	OEt	3.5	2.1
2	Et	n-Pr	OEt	5.2	2.4
3	Me	i-Pr	OEt	12.0	2.3
4	Me	n-Pr	OMe	8.7	1.9
5	Me	n-Pr	NH-Me	1.8	1.7

Table 1. Structure-activity relationships for compounds 1–5.

Claims

1. A compound of formula (I) wherein R₁ is C₁–C₃ alkyl, R₂ is n-propyl, and R₃ is selected from OMe, OEt, and NH-C₁–C₃-alkyl, or a pharmaceutically acceptable salt thereof.
2. The compound of claim 1, wherein R₁ is methyl and R₃ is OEt.